

E-Cell AI & Automation - Task 2

Knowledge Management & Semantic Retrieval System

- Enterprise Knowledge Management & Semantic Retrieval (RAG)
- Fully local inference with Ollama + ChromaDB
- GitHub: [KanakMalpani/Ecell-Projects](#)

Problem Statement

- Organizational docs hold critical operational knowledge
- Manual search is slow, error-prone, and inefficient
- Goal: ingest PDFs/text, index embeddings, run grounded Q&A via API

Five-Stage Pipeline

- Stage 1: Document ingestion & text segmentation (pdfplumber, chunking)
- Stage 2: Embedding generation & ChromaDB indexing (MiniLM-L6-v2)
- Stage 3: LLM inference & context orchestration (Ollama + reranker)
- Stage 4: Pipeline evaluation (CR, F, AR, L, QR)
- Stage 5: FastAPI deployment (POST /query with source metadata)

Three Pipeline Paths (Ollama)

- local_llm - vector search + Ollama (qwen2.5:7b)
- reranked_local - cross-encoder rerank + Ollama [RECOMMENDED]
- extractive - retrieval-only baseline (no LLM)

Evaluation Results

Pipeline	CR	F	AR	QR	L (ms)
local_llm	0.6241	0.6923	0.6871	0.7692	6052.9
reranked_local	0.6241	0.6807	0.7032	0.8462	6099.9
extractive	0.6241	0.6923	0.6871	0.7692	22.1

Anti-Hallucination Guardrails

- Context-only system prompt; mandatory source citations
- Similarity threshold (0.25) before LLM invocation
- Abstention phrase when context is insufficient
- Confidence derived from vector similarity scores
- Source metadata returned with every API response

Live Demo

- POST /query via FastAPI Swagger (127.0.0.1:8001/docs)
- Show sources[].similarity and source_file in responses
- Dynamic ingestion: backup_recovery_sop.pdf (7 docs, 43 chunks)
- Example: VPN lockout procedure, password policy, GDPR breach timeline

Deliverables & Submission

- Source code: AI and Automation/Task-2/ on GitHub
- System report: SYSTEM_REPORT.md + SYSTEM_REPORT.pdf
- Evaluation: reports/metrics_comparison.csv + .png
- This presentation: submission/Task-2-Presentation.pdf